Structure in Attitude Reports

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1 A hypothesis about 'explicitly say'

Consider

(1) The teacher explicitly said that four is even.

I suggest that (1)'s truth-conditions can be spelled out as follows:

(1*) The teacher produced an utterance u_1 composed of sub-utterances u_2 and u_3 (with the latter in the role of argument) such that u_3 suitably refers to 4, and u_2 suitably refers to *being even*.

This gives an appealing explanation of why (1) would be false if, e.g., the teacher only uttered the words 'three is odd'.

The word 'suitably' in (1*) is intended to stand for a context-sensitive restriction. This is motivated by evidence that attitude- and speech-reports in general, and 'explicitly say' reports in particular, are context-sensitive.

- For example, if the detective said 'Jack the Ripper did this crime', then it's plausible that 'The detective explicitly said that you did this crime' could be used either truly or falsely while addressing Jack the Ripper.

The method that turned (1) into (1*) can be generalized to a wide variety of expressions of the form 'S explicitly said that P'. For example, insofar as (1) can be spelled out as (1*), plausibly (2) can be spelled out as (2*)

(2) The teacher explicitly said that four is a multiple of two

(2*) The teacher produced an utterance u_1 composed of sub-utterances u_2 and u_3, where u_3 suitably refers to 4, and u_2 is composed in turn of sub-utterances u_4 and u_5, where u_5 suitably refers to 2, and u_4 suitably refers to *being a multiple of*.

One might worry that these truth conditions are too demanding even for 'explicitly say'. (What about sign language? Languages that don't have a binary branching structure?) But they provide a reasonable starting point.

2 The status of substitution

Substituting a proper name in an 'explicitly say' report for another proper name for the same object will make no difference to the truth conditions generated by this method:

(3) a. Lois explicitly said that Superman flies.
  b. Lois explicitly said that Clark flies.

Such substitutions can, however, affect which resolutions of the context-sensitive "suitability" restrictions are most *salient* or *natural*: e.g., referring to Kal-El as 'Clark' can bias us towards an interpretation where utterances of 'Superman' don't count as suitable. But these effets are defeasible, and I doubt that they involve special compositional mechanisms. (See **DorrTCAR** for more.)

Substitution of a *simple* predicate for another simple predicate that denotes the same property works just the same:

(4) a. She thinks that lawyers need a special license.
  b. She thinks that attorneys need a special license.

But the account gives a very different status to substitution of coreferential predicates when one or both is complex.

(5) a. She explictly said that Fermat's Last Theorem was a theorem of Peano Arithmetic.
  b. She explictly said that Fermat's Last Theorem was a theorem of Peano Arithmetic together with Fermat's Last Theorem.

(6) a. Brouwer explicltly said that the decimal expansion of π didn't both contain and not contain 100 consecutive 7s.
  b. Brouwer explicltly said that the decimal expansion of π either contained or did not contain 100 consecutive 7s.

It is not obvious that these are cases where the substituted predicates denote the same property. This is so according to Intensionalism (Bacon and Dorr 2024), but there are some (**SalmonAM**) who treat these apparent substitution-failures as a *reductio* of Intensionalism and other such coarse-grainedness views about the individuation of properties. But I don't think this is good, since there are other cases where substitution seems to fail, and the claim that the predicates denote the same property is plausible indepedent of any general coarse-grainedness principles:

(7) a. The teacher explicltly said that there is a tadpole in the bowl.
  b. The teacher explicltly said that there is a juvenile frog in the bowl.

(8) a. This line of the deduction explicltly says that every square is a four-sided polygon with equal sides and angles.
  b. This line of the deduction explicltly says that every square is a square.

(9) a. I explicltly said that the lecture would cover semigroups.
  b. I explicltly said that the lecture would cover associative magmas.

3 Structure-sensitivity for attitude verbs

If 'explicitly say' is structure-sensitive in the way our hypothesis suggests, it is plausible that this is also true for a wide array of speech and attitude verbs, including 'believe' (the standard paradigm). But it is not obvious how to adapt our theory of 'explicitly say' to turn it into a theory of 'believe'. I'll discuss four possibilities: what I care about here is their common structure.

I'll illustrate with the *that*-clauses 'that Swimmy is a tadpole' and 'that Swimmy is a juvenile frog'. I'll treat 'is' and 'a' as vacuous; 'Swimmy' as (uniquely, context-insensitively) denoting an individual Swimmy; 'tadpole' and 'frog' as denoting properties of individuals *tadpole* and *frog*; and 'juvenile' as denoting a property-modifying operation *juvenility*.[1]

3.1 Mentalese account of belief

(i) 'believes that Swimmy is a tadpole' expresses the property of being someone whose belief box (cf. Schiffer 1981) contains a Mentalese sentence (-type) consisting of a predicate t_1 applied to a name t_2, where t_2 suitably refers to Swimmy and t_1 suitably refers to *being a tadpole*.

(ii) 'Believes that Swimmy is a juvenile frog expresses the property of being someone whose belief box contains a sentence that consists of a predicate t_1 applied to a name t_2, where t_2 suitably refers to Swimmy, and t_1 consists in turn of a modifier t_3 applied to a predicate t_4, where t_3 suitably refers to *juvenility* and t_4 suitably refers to *being a frog*.

3.2 Descriptivist account of belief

We posit an underlying relation of *doxizing* between agents and states-of-affairs.

(i) 'Believes that Swimmy is a tadpole' expresses the property of doxizing some state of affairs of the form $\exists x^{e\to t} y^e . Xx \wedge Yy \wedge xy$, where Y is a suitable property of Swimmy and X is a suitable property of *being a tadpole*.

(ii) 'Believes that Swimmy is a juvenile frog' expresses the property of doxizing some state of affairs of the form $\exists x^{e\to t} y^e z^{(e\to t)\to e\to t} . Xx \wedge Yy \wedge Zz \wedge (zx)y$, where Y is a suitable property of Swimmy, X is a suitable property of *being a frog*, and Z is a suitable property of *juvenility*.

- We might or might not follow Hintikka in equating 'u doxises p' with 'p holds in every world that is doxastically possible for u'.
- But if we do, we should be wary of glossing 'w is doxastically possible with u' as 'w is compatible with everything u believes': taken at face value, this will have the undesirable consequence no worlds are doxastically possible for someone who believes that Ortcutt is a spy and Ortcutt is not a spy.

[1] Type $(e \to t) \to e \to t$, where t is the type of propositions

3.3 A Lewisian *de se* variant

We posit an underlying relation of *self-ascription* between agents and properties.

(i) 'Believes that Swimmy is a tadpole' expresses the property of self-ascribing some property of the form $\lambda v . \exists x^{e\to t} y^e . Xxv \wedge Yyv \wedge xy$, where Y is a suitable relation Swimmy stands in to u and X is a suitable relation *being a tadpole* stands in to u.

(ii) 'Believes that Swimmy is a juvenile frog expresses the property of self-ascribing some property some property of the form $\lambda v . \exists x^{e\to t} y^e z^{(e\to t)\to e\to t} . Xxv \wedge Yyv \wedge Zzv \wedge (zx)y$, where Y is a suitable relation Swimmy stands in to u, X is a suitable relation *being a frog* stands in to u, and Z is a suitable relation *juvenility* stands in to u.

3.4 A disjunctive variant

Suppose that in Quine's 1956 vignette, Ralph says to himself: 'Maybe that guy I saw on the beach isn't a spy, and maybe that pillar of the community I see on TV isn't a spy. But for sure, at least one of the two is a spy'. Does Ralph believe that Ortcutt is a spy?

Most views in the literature (following Quine (1956) and Kaplan (1968)) answer *no*, on the grounds that believing Ortcutt is a spy requires believing him to be a spy under some "guise", and there is no guise under which Ralph believes Ortcutt to be a spy. But I rather think yes!

We can tweak the desciptivist truth-conditions from subsection 3.2 or subsection 3.3 to get this by replacing 'some state of affairs/property...' with 'the disjunction of all states of affairs/properties...'.

4 A compositional implementation

The challenge: give a compositional explanation of such structure-sensitive truth-conditions. I'll describe a flexible system whose paramaters can be adjusted to fit all the views we have discussed.

- For each type σ, we have another type $\sigma^\dagger$. We could think of this as the type of 'senses' for type-σ things (though in many respects our theory will be quite unlike Frege's!)
- Attitude and speech predicates that combine with 'that' clauses require an argument of type $t^\dagger$, where one would naïvely expect t.
- To build a type-$t^\dagger$ denotation for a sentence out of ordinary words, we can

apply two families of silent operators

$$\mathbf{R}_\sigma : \sigma \to \sigma^\dagger$$
$$\mathbf{J}_{\sigma,\tau} : (\sigma \to \tau)^\dagger \to \sigma^\dagger \to \tau^\dagger$$

- **J** can be applied anywhere it makes sense, but **R** operators are only available for *non-compound* expressions.
- The **R** operators are the key source of context-sensitivity; when there are several, they create the potential for non-uniform interpretation.

For illustrative purposes, I'll focus first on the parameter-settings that deliver the Mentalese truth conditions from subsection 3.1. Where ε is the type of Mentalese expressions, we put:

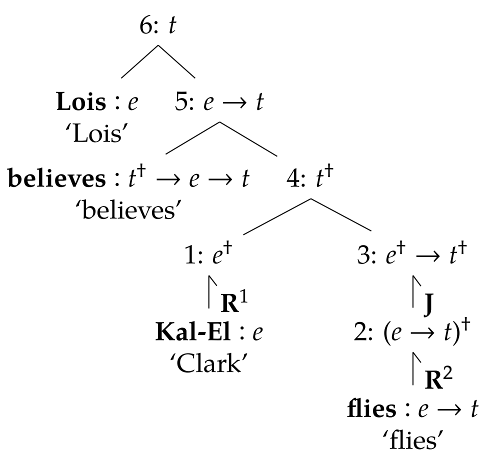
$$\begin{aligned} \sigma^\dagger &:= \varepsilon \to t \quad \text{(for any type } \sigma\text{!)} \\ \textbf{believes} &:= \lambda F^{\varepsilon\to t} u^e.\, \exists a^\varepsilon.\, Fa \wedge \textbf{bbox}\, a\, u \\ \mathbf{J}_{\sigma,\tau} &:= \lambda F^{\varepsilon\to t} G^{\varepsilon\to t} a^\varepsilon.\, \exists b^\varepsilon c^\varepsilon. Fb \wedge Gc \wedge \textbf{app}\, a\, b\, c \\ \mathbf{R}_\sigma &:= \lambda x^\sigma a^\varepsilon.\, a \text{ suitably refers to } x \end{aligned}$$

where **app** $a\,b\,c$ means 'a is the expression that results from combining b with c as argument', and **bbox** $a\,u$ means 'a is in u's belief box'.

Idea: a "sense determining x" is a property of Mentalese expressions that entails *referring to x*.

Let's work through some examples.

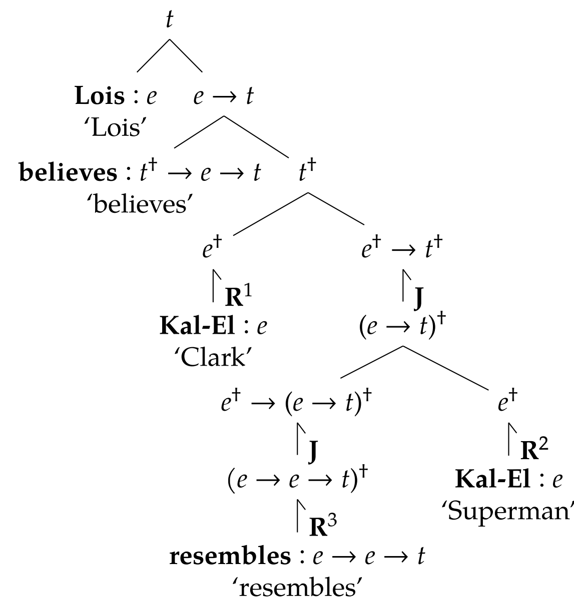
(10) Lois believes Clark flies.



Given the rule that **R** can only be applied to non-compound expressions, this is the only possible parse of (10).[2]

Node	*Denotation*
1	$\lambda c^\varepsilon.\, \mathbf{R}^1\, \textbf{Kal-El}\, c$
2	$\lambda b^\varepsilon.\, \mathbf{R}^2\, \textbf{flies}\, b$
3	$\lambda X^{\varepsilon\to t} a^\varepsilon.\, \exists b^\varepsilon c^\varepsilon. \mathbf{R}^2\, \textbf{flies}\, b \wedge Xc \wedge \textbf{app}\, a\, b\, c$
4	$\lambda a^\varepsilon. \exists b^\varepsilon c^\varepsilon. \mathbf{R}^2\, \textbf{flies}\, b \wedge \mathbf{R}^1\, \textbf{Kal-El}\, c \wedge \textbf{app}\, a\, b\, c$
5	$\lambda u^e. \exists a^\varepsilon b^\varepsilon c^\varepsilon. \mathbf{R}^2\, \textbf{flies}\, b \wedge \mathbf{R}^1\, \textbf{Kal-El}\, c \wedge \textbf{app}\, a\, b\, c \wedge \textbf{bbox}\, a\, u$
6	$\exists a^\varepsilon b^\varepsilon c^\varepsilon. \mathbf{R}^2\, \textbf{flies}\, b \wedge \mathbf{R}^1\, \textbf{Kal-El}\, c \wedge \textbf{app}\, a\, b\, c \wedge \textbf{bbox}\, a\, \textbf{Lois}$

(11) Lois believes that Clark resembles Superman.



It now matters a lot whether $\mathbf{R}^1$ and $\mathbf{R}^2$ are the same or different relations.

Truth condition when $\mathbf{R}^2 = \mathbf{R}^1$: Lois's belief box contains a sentence *Fab*, where F suitably$_3$ refers to *resemblance*, and a and b both suitably$_1$ refer to Kal-El.

General truth condition: Lois belief box contains a sentence *Fab*, where F suitably refers to *resemblance*, a suitably$_2$ refers to Kal-El, and b suitably$_1$ refers to Kal-El.

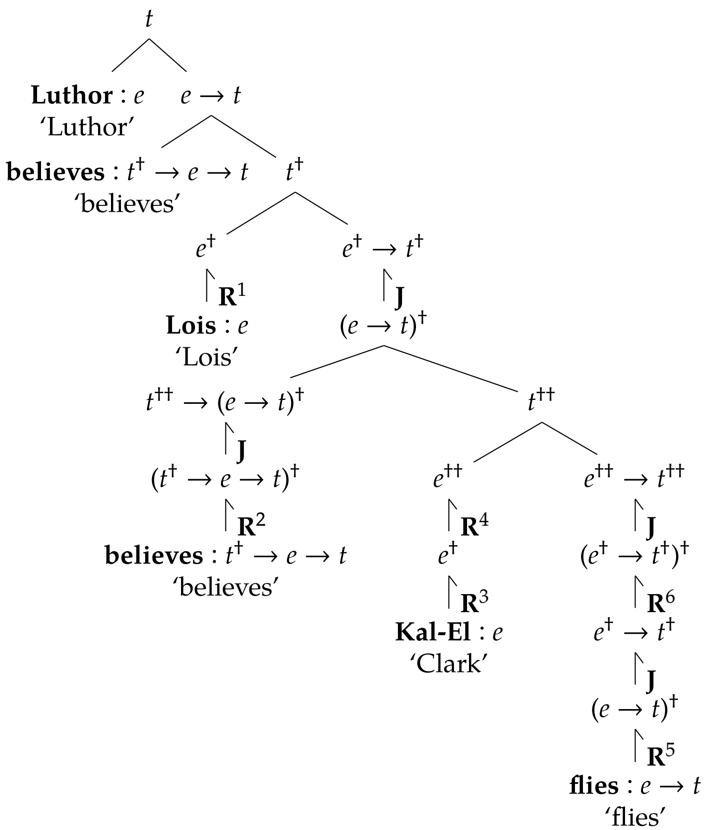
- For the former to be false, we either have to be in a weird context where

[2] So, we aren't just treating $t^\dagger$ as an abbreviation for something else, but using the dagger to determine what's an allowed parse. In a more elaborate system, this work could be done by a system of 'category labels' separate from semantic types.

neither of Lois's Mentalese names for Kal-El counts as "suitable", or else Lois must be bad at thinking about the symmetry of resemblance. By contrast, given the choice of names, it's natural to think that Lois's Mentalese version of 'Clark' suitably$_1$ but not suitably$_2$ refer so Kal-El, while her Mentalese version of 'Superman' does so suitably$_2$ but not suitably$_1$. Then the latter will be false so long as her Mentalese version of 'Clark resembles Superman' isn't in her belief box.

Our approach thus avoids a major problem that sentences like (11) raise for contextualists who want to pin all relevant context-sensitivity on 'believe' (Goodman and Lederman 2021).

(12) Luthor believes Lois believes Clark flies.



Truth-condition: Luthor's belief-box contains a sentence of the form $B(FC)a$, where: B suitably$_2$ refers to **believes** : $t^\dagger \to e \to t$; a suitably$_1$ refers to Lois; C suitably$_4$ refers to *suitably$_3$ referring to Kal-El*; and F suitably$_6$ refers to $\mathbf{J}$(*suitably$_5$ referring to flying*).[3]

Particularly interesting is the potential for non-uniform interpretions of $\mathbf{R}^3 : e \to e^\dagger$ and $\mathbf{R}^4 : e^\dagger \to e^{\dagger\dagger}$, such as the following:

- $\mathbf{R}^3$: a rather undemanding relation that Lois's Mentalese versions of 'Superman' and 'Clark' both bear to Kal-El.
- $\mathbf{R}^4$: a rather demanding relation that none of Luthor's complex predicates of the form 'does such-and-such to Superman' bear to *bearing* $\mathbf{R}^3$ *to Kal-El*, although certain predicates of the form 'does such-and-such to Clark' do bear it to that property.

This gives an interpretation where the sentence is demanding as regards how Luthor is thinking of Kal-El, but undemanding as regards the relation he takes Lois to bear to Kal-El.

5 Other ways of filling in the parameters

5.1 Structure-sensitive 'say'

As above, but with the type ν of events in the role of ε, and

$$\textbf{say} := \lambda F^{t^\dagger} e^\nu . Fe$$

(Of course getting denotations for sentences out of this will need some compositional implementation of event semantics, perhaps as in....)

5.2 Descriptivist theory of belief

$$\begin{aligned}
\sigma^\dagger &:= (\sigma \to t) \to t \\
\textbf{believes} &:= \lambda F^{t^\dagger} u^e . \exists X^{t \to t} . FX \wedge \textbf{dox}\, u\, (\exists p^t . p \wedge Xp) \\
\mathbf{J}_{\sigma,\tau} &:= \lambda F^{(\sigma\to\tau)^\dagger} G^{\sigma^\dagger} X^{\tau\to t} . \exists Y^{(\sigma\to\tau)\to t} Z^{\sigma \to t} . \\
&\qquad\qquad FY \wedge GZ \wedge X = (\lambda x^\tau . \exists y^{\sigma\to\tau} z^\sigma . Yy \wedge Zz \wedge x = yz) \\
\mathbf{R}_\sigma &:= \lambda x^\sigma Y^{\sigma \to t} . Yx \wedge Y \text{ is a suitable property}
\end{aligned}$$

where $\textbf{dox} : e \to t \to t$ represents the posited underlying *doxizing* relation.

So, going back to our tree for (10), the numbered nodes receive the following denotations (where $\mathbf{S}_1 : (e \to t) \to t$ and $\mathbf{S}_2 : ((e \to t) \to t) \to t$ are the operative interpretations of 'suitable')

[3] That is: to *being a property of names and a sentence such that the sentence is the result of applying a predicate that suitably$_6$ refers to flying to a name with that property.*

1 $\lambda Z^{e\to t}.\, Z\,\textbf{Kal-El} \wedge \textbf{S}_1\, Z$

2 $\lambda Y^{(e\to t)\to t}.\, Y\,\textbf{flies} \wedge \textbf{S}_2\, Y$

3 $\lambda G^{e^\dagger} X^{t\to t}.\exists Y^{(e\to t)\to t} Z^{e\to t}. Y\,\textbf{flies} \wedge \textbf{S}_2\, Y \wedge GZ$
$\wedge\, X = (\lambda p^t.\, \exists y^{e\to t} z^e. Yy \wedge Zz \wedge p = yz)$

4 $\lambda X^{t\to t}.\exists Y^{(e\to t)\to t} Z^{e\to t}. Y\,\textbf{flies} \wedge \textbf{S}_2\, Y \wedge Z\,\textbf{Kal-El} \wedge \textbf{S}_1\, Z$
$\wedge\, X = (\lambda p^t.\, \exists y^{e\to t} z^e. Yy \wedge Zz \wedge p = yz)$

5 $\lambda u^e.\exists X^{t\to t} Y^{(e\to t)\to t} Z^{e\to t}. Y\,\textbf{flies} \wedge \textbf{S}_2\, Y \wedge Z\,\textbf{Kal-El} \wedge \textbf{S}_1\, Z$
$\wedge\, X = (\lambda p^t.\, \exists y^{e\to t} z^e. Yy \wedge Zz \wedge p = yz) \wedge \textbf{dox}\, u\, (\exists p^t. p \wedge Xp)$

6 $\exists X^{t\to t} Y^{(e\to t)\to t} Z^{e\to t}. Y\,\textbf{flies} \wedge \textbf{S}_2\, Y \wedge Z\,\textbf{Kal-El} \wedge \textbf{S}_1\, Z$
$\wedge\, X = (\lambda p^t.\, \exists y^{e\to t} z^e. Yy \wedge Zz \wedge p = yz) \wedge \textbf{dox}\,\textbf{Lois}(\exists p^t. p \wedge Xp)$

Given Intensionalism, 6 can be simplified to

$$\exists Y^{(e\to t)\to t} Z^{e\to t}. Y\,\textbf{flies} \wedge \textbf{S}_2\, Y \wedge Z\,\textbf{Kal-El} \wedge \textbf{S}_1\, Z \wedge \textbf{dox}\,\textbf{Lois}(\exists y^{e\to t} z^e. Yy \wedge Zz \wedge yz)$$

These truth-conditions are defensible only if we take "suitable" always to require *necessarily having at most one instance*.

5.3 Lewisian variant of descriptivism

The role of **dox** is now played by a relation **self-ascribes**, of type $e \to (e \to t) \to t$. We carry around an extra type-e argument everywhere:

$$\sigma^\dagger := (\sigma \to e \to t) \to e \to t$$

$$\textbf{believes} := \lambda F^{t^\dagger} u^e.\exists X^{t\to e\to t}. FXu \wedge \textbf{self-ascribes}\, u\, (\lambda v^e.\exists p^t. p \wedge Xpv)$$

$$\textbf{J}_{\sigma,\tau} := \lambda F^{(\sigma\to\tau)^\dagger} G^{\sigma^\dagger} X^{\tau\to e\to t} u^e.\exists Y^{(\sigma\to\tau)\to e\to t} Z^{\sigma\to e\to t}. FYu \wedge GZu \wedge X = (\lambda x^\tau v^e.\exists y^{\sigma\to\tau} z^\sigma. Yyv \wedge Zzv \wedge x = yz)$$

$$\textbf{R}_\sigma := \lambda x^\sigma Y^{\sigma\to e\to t} u^e. Yxu \wedge Y \text{ is a suitable relation}$$

So, (10) gets the following truth condition (assuming Intensionalism for simplicity):

(13) There is a suitable relation Y that *flies* bears to Lois, and a suitable relation Z that Kal-El bears to Lois, such that Lois self-ascribes $\lambda v^e.\exists y^{e\to t} z^e. Yyv \wedge Zzv \wedge yz$.

5.4 Disjunctive variant of descriptivism

Same as for the standard form of descriptiviem, except that

$$\textbf{believes} := \lambda F^{t^\dagger} u^e.\, \textbf{dox}\, u\, (\bigvee_{X:FX} \exists p^t. p \wedge Xp)$$

So, (10) gets the following truth condition:

(14) Lois doxizes the disjunction of all propositions of the form $\exists p^t. p \wedge \exists y^{e\to t} z^e. Yy \wedge Zz \wedge p = yz$, where Y is a suitable property of flying and Z is a suitable property of Kal-El.

Assuming doxizing is closed under entailment, this is weaker than the standard descriptive truth-condition.

6 Quantification over the objects of attitudes

This argument feels valid:

(15) a. John believes everything that the Pope believes.
b. The Pope believes Swimmy is a juvenile frog.
c. Therefore, John believes Swimmy is a juvenile frog.

We can straightforwardly validate this, by interpreting the quantification as into type $t^\dagger$.[4]

- This is crucial advantage of my approach over various other ways of generating the same truth conditions for 'S believes that P' (e.g. those of Cresswell and von Stechow (1982), Cresswell (1985), Heim (1994), and Percus and Sauerland (2003)). These approaches all treat something in these sentences (maybe 'believe', maybe 'that', maybe some covert eleemnt) as taking types that vary depending on the structure of the prejacent P. But this makes a uniform treatment of quantification impossible.

But what about 'Everything the Pope believes is true/false'? What could 'true' and 'false' express, in type $t^\dagger \to t$?

Key desideratum: each of the various $t^\dagger$ things that can be denoted by 'Clark flies' is such that, necessariliy, it is true iff Clark flies.

Mentalese suggestion: 'True' expresses $\lambda X^{\varepsilon\to t}.\, \exists p.\, p \wedge X \leq \lambda a^\varepsilon. a$ refers to p.[5]

[4] We could also interpret (15a) using type-t quantification, applying $\textbf{R}$ to lift the bound variable from t to $t^\dagger$: we could think of this as meaning 'For every state of affairs that the Pope believes to obtains, John also believes it to obtain'. But that interpretation of (15a) would invalidate the argument.

[5] Here, $\leq$ denotes *entailment* among properties/relations/states of affairs. We could give a parallel account of 'false' with a negation in front of the first p; 'necessary' with a box in front of the p; and so on.

- For example: 'That Clark flies is true' expresses

(16) $\exists p^t.\, p \wedge (\lambda a^\varepsilon.\exists b^\varepsilon c^\varepsilon.\mathbf{R}_1\,\mathbf{flies}\,b \wedge \mathbf{R}_2\,\mathbf{Kal\text{-}El}\,c \wedge \mathbf{app}\,a\,b\,c) \leq (\lambda a^\varepsilon.\,\mathbf{refers}\,pa)$

- (16) follows from **flies Kal-El**, if we assume that $\mathbf{R}_1$ and $\mathbf{R}_2$ both entail **refers**, and the compositional axiom $\mathbf{app}\,a\,b\,c \wedge \mathbf{refers}\,X\,b \wedge \mathbf{refers}\,y\,c \rightarrow \mathbf{refers}\,(Xy)\,a$.
- Unfortunately, I can see no plausible general laws which would license the inference from (16) to **flies Kal-El**. If we allow for reference to be non-unique (as I think we should), it's hard to rule out the scenario where necessarily any utterance composed of parts referring to Kal-El and flying also refers to some *other* state-of-affairs. If we don't, then to avoid paradox we have to allow for some things that are impossible to refer to, which also breaks the proposal.
- To avoid this problem, I think the best option is to modify the Mentalese strategy by taking $\sigma^\dagger$ to be the product type $\sigma \times (\varepsilon \rightarrow t)$. Then 'true' can just express $\lambda x^{t^\dagger}.\pi_0 x$.

Descriptivist suggestion: 'True' expresses $\lambda F^{(t\rightarrow t)\rightarrow t}.\,\exists p^t.\, p \wedge F \leq \lambda X^{t\rightarrow t}.Xp$.[6]

- So, 'It's true that Clark flies' expresses something equivalent to

$$\exists p^t.\, p \wedge \Box\forall Y^{(e\rightarrow t)\rightarrow t} Z^{e\rightarrow t}.\, Y\,\mathbf{flies} \wedge \mathbf{S}_2\,Y \wedge Z\,\mathbf{Kal\text{-}El} \wedge \mathbf{S}_1\,Z \\ \rightarrow \exists y^{e\rightarrow t} z^e.Yy \wedge Zz \wedge p = yz$$

- This follows immediately from **flies Kal-El**, given standard modal logic.
- It also implies **flies Kal-El**, given certain posits which seem defensible. The following would suffice: whenever $\mathbf{S} : (\sigma \rightarrow t) \rightarrow t$ is an allowable interpretation of 'suiitable', (i) $\mathbf{S}$ entails being such that one cannot be multiply instantiated ($\Box\forall X^{\sigma\rightarrow t}.\mathbf{S}X \rightarrow \Box\forall y^\sigma z^\sigma.Xy \wedge Xz \rightarrow y = z$) and (ii) Everything has some $\mathbf{S}$ property ($\forall x^\sigma.\exists X^{\sigma\rightarrow t}.\mathbf{S}X \wedge Xx$).

[6] *Lewisian variant:* $\lambda R^{t^\dagger}.\,\exists p^t.\, p \wedge R \leq \lambda S^{t\rightarrow e\rightarrow t} u^e.Spu$.

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